



# BCA 3rd Semester - Web Development (HTML, CSS)



## UNIT – 1

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### (a) Introduction to Web Development



#### i) Basics of Internet and Web

##### **World Wide Web (WWW)**

The World Wide Web is a global information system that enables users to access and share information through interconnected hypertext documents. It operates on top of the Internet infrastructure and provides a user-friendly interface for accessing resources.

##### **Key Components:**

- **Hypertext Documents:** Web pages containing text, images, multimedia, and hyperlinks
- **Uniform Resource Locators (URLs):** Unique addresses for web resources
- **Hypertext Transfer Protocol (HTTP/HTTPS):** Communication protocol for data exchange
- **Domain Name System (DNS):** Translates domain names to IP addresses

##### **Web Browsers**

Web browsers are software applications that retrieve, present, and navigate information resources on the World Wide Web. They interpret HTML, CSS, and JavaScript to display web pages.

### **Popular Web Browsers:**

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari
- Opera

### **Browser Functions:**

- **Rendering Engine:** Interprets HTML, CSS, and JavaScript
- **JavaScript Engine:** Executes client-side scripts
- **User Interface:** Provides navigation tools and user controls
- **Security Features:** Protects against malicious content
- **Cache Management:** Stores frequently accessed resources locally

### **Web Servers**

Web servers are computer systems that host websites and serve web pages to clients upon request. They process HTTP requests and deliver appropriate responses.

### **Server Types:**

- **Apache HTTP Server:** Open-source, widely used

- **Nginx:** High-performance, lightweight
- **Microsoft IIS:** Windows-based server
- **LiteSpeed:** Commercial high-performance server

### Server Functions:

- **Request Processing:** Handles incoming HTTP requests
- **Content Delivery:** Serves static and dynamic content
- **Security Management:** Implements access control and encryption
- **Load Balancing:** Distributes traffic across multiple servers
- **Logging:** Records server activities and errors

### ii) Client-side vs Server-side Scripting

| Aspect             | Client-side Scripting       | Server-side Scripting                    |
|--------------------|-----------------------------|------------------------------------------|
| Execution Location | User's browser              | Web server                               |
| Languages          | JavaScript, HTML, CSS       | PHP, Python, Java, C#, Ruby              |
| Processing Speed   | Fast (local execution)      | Depends on server performance            |
| Security           | Less secure (code visible)  | More secure (code hidden)                |
| User Interaction   | Real-time, interactive      | Requires server communication            |
| Examples           | Form validation, animations | Database operations, user authentication |

### Client-side Scripting Characteristics:

- **Immediate Response:** No server round-trip required
- **Reduced Server Load:** Processing done on user's machine
- **Browser Dependency:** Functionality may vary across browsers
- **Limited Security:** Code accessible to users
- **Offline Capability:** Can function without internet connection

### **Server-side Scripting Characteristics:**

- **Enhanced Security:** Source code not visible to users
- **Database Integration:** Direct access to server databases
- **Cross-browser Compatibility:** Output is standard HTML
- **Resource Intensive:** Requires server processing power
- **Real-time Data:** Access to current server information



### **iii) Structure of a Web Page**

A typical web page follows a hierarchical structure that defines its organization and presentation:

### **Basic Structure Components:**

#### **1. Document Type Declaration (DOCTYPE)**

- Specifies HTML version
- Ensures browser compatibility
- Triggers standards mode rendering

#### **2. HTML Root Element**

- Container for all page content
- Defines document language

- Establishes document scope

### 3. Head Section

- Contains metadata about the document
- Includes title, character encoding, stylesheets
- Not visible to users directly

### 4. Body Section

- Contains visible page content
- Includes text, images, links, forms
- Where user interaction occurs

### Semantic Structure Elements:

- **Header:** Page or section introduction
- **Navigation:** Site navigation links
- **Main:** Primary content area
- **Article:** Independent, self-contained content
- **Section:** Thematic grouping of content
- **Aside:** Sidebar or supplementary content
- **Footer:** Page or section conclusion

## iv) Introduction to HTML

### HTML Syntax

HTML (HyperText Markup Language) is the standard markup language for creating web pages. It uses a tag-based syntax to structure and present content.

## Syntax Rules:

- **Case Insensitive:** Tags can be uppercase or lowercase (lowercase recommended)
- **Nested Structure:** Tags must be properly nested and closed
- **Attribute Syntax:** Attributes provide additional information about elements
- **Whitespace Handling:** Multiple spaces and line breaks are treated as single space

## HTML Elements

Elements are the building blocks of HTML documents, consisting of start tags, content, and end tags.

### Element Categories:

- **Block Elements:** Take full width, start on new line (div, p, h1-h6)
- **Inline Elements:** Take only necessary space (span, a, strong, em)
- **Empty Elements:** Self-closing, no content (img, br, hr, input)
- **Container Elements:** Hold other elements (html, body, head)

## HTML Tags

Tags are keywords enclosed in angle brackets that define elements and their properties.

### Tag Types:

- **Opening Tags:** Mark the beginning of an element

- **Closing Tags:** Mark the end of an element (with forward slash)
  - **Self-closing Tags:** Complete elements without separate closing tags
  - **Conditional Tags:** Used for browser-specific content
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## (b) HTML Fundamentals



### Text Formatting Tags

HTML provides various tags for formatting text appearance and emphasizing content meaning.

#### Basic Text Formatting:

- **Bold Text:** Makes text appear bold for emphasis
- **Italic Text:** Makes text appear slanted for emphasis
- **Underlined Text:** Adds underline to text
- **Strikethrough Text:** Shows text with line through it
- **Subscript Text:** Displays text below baseline
- **Superscript Text:** Displays text above baseline

#### Semantic Text Elements:

- **Strong Importance:** Indicates strong importance or urgency
- **Emphasis:** Provides emphasis or stress
- **Mark/Highlight:** Highlights relevant text
- **Small Text:** Represents side comments or fine print
- **Deleted Text:** Shows removed content

- **Inserted Text:** Shows added content

## Preformatted Text:

- **Preformatted:** Preserves whitespace and formatting
- **Code:** Represents computer code
- **Keyboard Input:** Shows keyboard input
- **Sample Output:** Displays sample program output
- **Variable:** Represents programming variables



## Lists in HTML

| List Type       | Purpose               | Common Use Cases                    |
|-----------------|-----------------------|-------------------------------------|
| Ordered List    | Numbered sequence     | Step-by-step instructions, rankings |
| Unordered List  | Bulleted items        | Navigation menus, feature lists     |
| Definition List | Term-definition pairs | Glossaries, FAQs                    |

### Ordered Lists:

Ordered lists display items in numerical sequence, indicating hierarchy or sequence importance.

### Characteristics:

- **Automatic Numbering:** Browser generates numbers automatically
- **Customizable Numbering:** Can use different number types (1, A, a, I, i)
- **Start Attribute:** Can begin numbering from specific value
- **Nested Lists:** Support multiple levels of numbering



- **Semantic Meaning:** Indicates order or priority

## Unordered Lists:

Unordered lists display items with bullet points, indicating related but not sequential items.

### Characteristics:

- **Bullet Points:** Default circular bullets
- **Customizable Bullets:** Can change bullet style (disc, circle, square)
- **Nested Structure:** Support multiple indentation levels
- **Flexible Styling:** Easy to customize with CSS
- **Semantic Grouping:** Groups related items together

## Definition Lists:

Definition lists present term-definition pairs, ideal for glossaries and explanations.

### Components:

- **Definition List Container:** Wraps entire list
- **Definition Term:** The term being defined
- **Definition Description:** The explanation or definition
- **Multiple Definitions:** Single term can have multiple definitions
- **Structured Format:** Clear separation between terms and definitions

## Hyperlinks

Hyperlinks are the foundation of web navigation, connecting different resources and enabling user interaction.

## **Internal Links:**

Internal links connect to resources within the same website or document.

### **Types of Internal Links:**

- **Page Links:** Connect to other pages in same site
- **Anchor Links:** Jump to specific sections within same page
- **Relative Links:** Use relative paths from current location
- **Fragment Links:** Link to specific elements using IDs
- **Directory Links:** Navigate through site structure

### **Benefits:**

- **Improved Navigation:** Easy movement within site
- **Better User Experience:** Quick access to related content
- **SEO Benefits:** Helps search engines understand site structure
- **Reduced Load Time:** No external server requests
- **Consistent Experience:** Maintains site design and functionality

## **External Links:**

External links connect to resources on different websites or domains.

### **Characteristics:**

- **Absolute URLs:** Complete web addresses including protocol

- **Target Attribute:** Can open in new window/tab
- **Security Considerations:** Should include rel="noopener" for security
- **Link Validation:** Should be checked regularly for broken links
- **User Indication:** Often styled differently to show external nature



## Images and Image Maps

### Images in HTML:

Images enhance web pages by providing visual content and supporting text information.

### Image Considerations:

- **File Formats:** JPEG, PNG, GIF, SVG, WebP
- **Optimization:** Balance quality and file size
- **Responsive Design:** Adapt to different screen sizes
- **Accessibility:** Provide alternative text descriptions
- **Loading Performance:** Consider lazy loading for large images

### Image Attributes:

- **Source (src):** Specifies image file location
- **Alternative Text (alt):** Describes image for accessibility
- **Width/Height:** Control image dimensions
- **Title:** Provides tooltip text
- **Loading:** Controls when image loads (lazy/eager)

### Image Maps:

Image maps allow different areas of an image to be clickable links.

### Image Map Components:

- **Image Element:** The base image
- **Map Element:** Defines clickable areas
- **Area Elements:** Specify individual clickable regions
- **Coordinate System:** Defines precise clickable areas
- **Shape Types:** Rectangle, circle, polygon

### Use Cases:

- **Navigation Graphics:** Interactive site maps
- **Product Catalogs:** Clickable product images
- **Educational Materials:** Interactive diagrams
- **Geographic Maps:** Clickable regions or locations
- **Architectural Plans:** Interactive building layouts



### Tables

Tables organize data in rows and columns, providing structured presentation of information.

### Table Structure:

HTML tables use a hierarchical structure to organize content systematically.

### Table Elements:

- **Table Container:** Wraps entire table structure

- **Table Header:** Contains column headings
- **Table Body:** Contains main data rows
- **Table Footer:** Contains summary or totals
- **Table Rows:** Horizontal data containers
- **Table Cells:** Individual data containers

## Creating Tables:

Tables require careful planning to ensure proper structure and accessibility.

### Table Planning:

- **Data Organization:** Determine rows and columns needed
- **Header Structure:** Identify column and row headers
- **Accessibility:** Ensure screen readers can navigate
- **Responsive Design:** Consider mobile viewing
- **Semantic Markup:** Use appropriate table elements

### Merging Rows/Columns:

Table cells can span multiple rows or columns for complex layouts.

### Spanning Techniques:

- **Column Span:** Cell extends across multiple columns
- **Row Span:** Cell extends across multiple rows
- **Combined Spanning:** Cell spans both rows and columns
- **Accessibility Impact:** May affect screen reader navigation
- **Layout Complexity:** Requires careful planning and testing

## **Forms**

Forms enable user interaction and data collection on web pages.

### **Input Elements:**

Forms contain various input elements for different types of data collection.

### **Common Input Types:**

- **Text Input:** Single-line text entry
- **Password Input:** Hidden text entry
- **Email Input:** Email address validation
- **Number Input:** Numeric value entry
- **Date Input:** Date picker interface
- **File Input:** File upload functionality
- **Checkbox:** Multiple selection options
- **Radio Button:** Single selection from group
- **Select Dropdown:** Dropdown menu options
- **Textarea:** Multi-line text entry

### **Form Attributes:**

Form attributes control how forms behave and process data.

### **Essential Attributes:**

- **Action:** Specifies where form data is sent
- **Method:** Defines how data is transmitted (GET/POST)

- **Enctype:** Specifies encoding type for file uploads
- **Target:** Determines where response is displayed
- **Autocomplete:** Controls browser autocomplete behavior
- **Novalidate:** Disables browser validation

## Form Validation Basics:

Form validation ensures data quality and improves user experience.

### Validation Types:

- **Required Fields:** Ensures essential data is provided
- **Format Validation:** Checks data format (email, phone)
- **Length Validation:** Controls minimum/maximum input length
- **Pattern Validation:** Uses regular expressions for complex rules
- **Custom Validation:** JavaScript-based validation logic
- **Server-side Validation:** Backend validation for security

### Validation Benefits:

- **Data Quality:** Ensures accurate information collection
- **User Experience:** Provides immediate feedback
- **Security:** Prevents malicious input
- **Processing Efficiency:** Reduces server-side errors
- **Accessibility:** Helps users understand requirements

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## Key Learning Objectives

By completing this unit, students will understand:

- ✓ **Web Development Fundamentals:** Core concepts of internet, web, and development approaches
  - ✓ **HTML Structure:** Proper document structure and semantic markup
  - ✓ **Content Organization:** Effective use of lists, tables, and formatting
  - ✓ **User Interaction:** Creating forms and interactive elements
  - ✓ **Navigation:** Implementing internal and external links
  - ✓ **Media Integration:** Including images and multimedia content
  - ✓ **Best Practices:** Writing clean, accessible, and maintainable HTML code
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## Next Steps

This unit provides the foundation for web development. The next units will cover:

- Advanced HTML features and HTML5 elements
  - CSS fundamentals and styling techniques
  - Responsive web design principles
  - JavaScript basics and interactivity
  - Modern web development practices
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*Happy Learning!* 🎓